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Jia et al.

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(54) **MULTI-USER MIMO SYSTEMS AND METHODS**

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H04L 1/02 (2006.01)
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CPC **H04W 72/542** (2023.01); **H04B 7/0452** (2013.01); **H04B 7/0478** (2013.01);
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CPC .. H04B 7/0425; H04B 7/0478; H04B 7/0632; H04B 7/073; H04B 7/0452;
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(56) **References Cited**

U.S. PATENT DOCUMENTS

7,103,325 B1 9/2006 Jia
7,792,227 B2 9/2010 Moorti
(Continued)

FOREIGN PATENT DOCUMENTS

WO 2003085875 A1 10/2003

OTHER PUBLICATIONS

3GPP TS 36.212 V8.8.0; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and Channel Coding; Release 8; Dec. 2009; 60 pages.

(Continued)

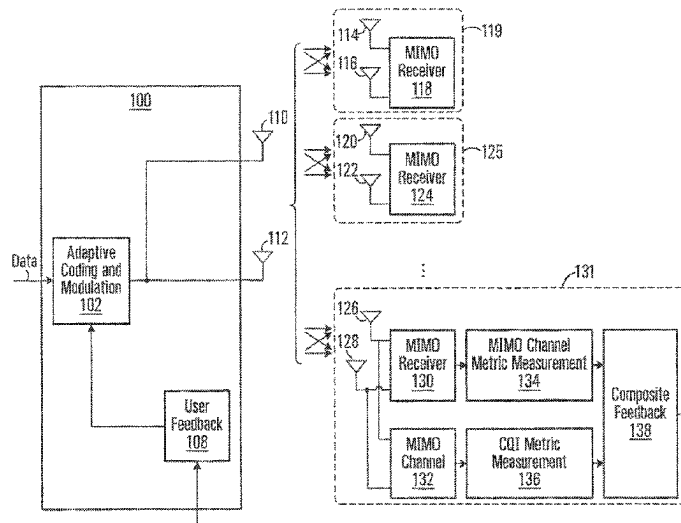
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(57) **ABSTRACT**

A method and system are provided for scheduling data transmission in a Multiple-Input Multiple-Output (MIMO) system. The MIMO system may comprise at least one MIMO transmitter and at least one MIMO receiver. Feedback from one or more receivers may be used by a transmitter to improve quality, capacity, and scheduling in MIMO communication systems. The method may include generating or receiving information pertaining to a MIMO channel metric and information pertaining to a Channel Quality Indicator (CQI) in respect of a transmitted signal; and sending a next transmission to a receiver using a MIMO mode selected in accordance with the information pertaining to the MIMO channel metric, and an adaptive coding and modulation selected in accordance with the information pertaining to the CQI.

20 Claims, 8 Drawing Sheets



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continuation of application No. 17/097,907, filed on Nov. 13, 2020, now Pat. No. 11,291,017, which is a continuation of application No. 16/384,478, filed on Apr. 15, 2019, now Pat. No. 10,856,305, which is a continuation of application No. 15/881,575, filed on Jan. 26, 2018, now Pat. No. 10,306,658, which is a continuation of application No. 15/360,600, filed on Nov. 23, 2016, now Pat. No. 9,918,328, which is a continuation of application No. 14/832,758, filed on Aug. 21, 2015, now Pat. No. 9,538,408, which is a continuation of application No. 13/971,534, filed on Aug. 20, 2013, now Pat. No. 9,301,174, which is a continuation of application No. 13/608,234, filed on Sep. 10, 2012, now Pat. No. 8,611,454, which is a continuation of application No. 13/251,394, filed on Oct. 3, 2011, now Pat. No. 8,284,852, which is a continuation of application No. 12/089,938, filed as application No. PCT/CA2006/001665 on Oct. 12, 2006, now Pat. No. 8,054,898.

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H04W 24/08 (2009.01)
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 USPC 375/267, 280, 262, 130, 148, 295, 347, 375/358; 370/319, 326, 329, 332
 See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,054,898 B2 11/2011 Tong
 8,249,518 B2 8/2012 Grant
 8,284,852 B2 10/2012 Tong
 8,599,945 B2 12/2013 Sampath
 8,611,454 B2 12/2013 Tong
 9,301,174 B2 3/2016 Tong
 9,313,553 B2 4/2016 Sridhar
 9,538,408 B2 1/2017 Tong
 9,918,328 B2 3/2018 Tong
 10,306,658 B2 5/2019 Tong
 10,856,305 B2 12/2020 Tong
 11,291,017 B2 3/2022 Tong

11,595,089 B2 2/2023 Ramireddy et al.
 11,765,763 B2 9/2023 Jia et al.
 2002/0196752 A1 12/2002 Attar
 2003/0058881 A1 3/2003 Wu
 2004/0162098 A1 8/2004 Wei
 2005/0025254 A1 2/2005 Awad
 2005/0043031 A1 2/2005 Cho
 2005/0085195 A1 4/2005 Tong
 2005/0147177 A1 7/2005 Seo
 2005/0181739 A1 8/2005 Krasny
 2005/0201296 A1 9/2005 Vannithamby
 2006/0013328 A1 1/2006 Zhang
 2006/0023745 A1 2/2006 Koo
 2006/0039312 A1 2/2006 Walton
 2006/0072508 A1 4/2006 Zou
 2006/0256843 A1 11/2006 Grant
 2007/0274253 A1 11/2007 Zhang
 2008/0132281 A1 6/2008 Kim
 2011/0150117 A1 6/2011 Zhou
 2013/0028230 A1 1/2013 Borran
 2016/0100401 A1* 4/2016 Xiong H04L 25/0242
 455/450
 2017/0170976 A1 6/2017 Ling
 2020/0220696 A1* 7/2020 Nammi H04L 1/0026

OTHER PUBLICATIONS

3GPP TS 36.213 V8.8.0; 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA); Physical Layer Procedures; Release 8; Sep. 2009; 77 pages.
 Office Action dated Jan. 7, 2011; U.S. Appl. No. 12/089,938, filed Apr. 11, 2008; 9 pages.
 Notice of Allowance dated Jun. 27, 2011; U.S. Appl. No. 12/089,938, filed Apr. 11, 2008; 7 pages.
 Office Action dated Feb. 1, 2012; U.S. Appl. No. 13/251,394, filed Oct. 3, 2011; 10 pages.
 Notice of Allowance dated Jun. 11, 2012; U.S. Appl. No. 13/251,394, filed Oct. 3, 2011; 8 pages.
 Office Action dated Mar. 5, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 6 pages.
 Office Action dated May 30, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 7 pages.
 Notice of Allowance dated Sep. 13, 2013; U.S. Appl. No. 13/608,234, filed Sep. 10, 2012; 8 pages.
 Office Action dated Aug. 28, 2014; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 7 pages.
 Office Action dated Nov. 17, 2014; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 5 pages.
 Notice of Allowance dated Feb. 11, 2015; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 7 pages.
 Notice of Allowance dated Aug. 14, 2015; U.S. Appl. No. 13/971,534, filed Aug. 20, 2013; 12 pages.
 Office Action dated Feb. 19, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 16 pages.
 Final Office Action dated Apr. 29, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 13 pages.
 Advisory Action dated Aug. 2, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 2 pages.
 Notice of Allowance dated Sep. 15, 2016; U.S. Appl. No. 14/832,758, filed Aug. 21, 2015; 8 pages.
 Office Action dated Sep. 15, 2017; U.S. Appl. No. 15/360,600, filed Nov. 23, 2016; 28 pages.
 Notice of Allowance dated Dec. 12, 2017; U.S. Appl. No. 15/360,600, filed Nov. 23, 2016; 8 pages.
 Office Action dated Sep. 10, 2018; U.S. Appl. No. 15/881,575, filed Jan. 26, 2018; 22 pages.
 Notice of Allowance dated Feb. 13, 2019; U.S. Appl. No. 15/881,575, filed Jan. 26, 2018; 8 pages.
 Office Action dated Oct. 24, 2019; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 21 pages.
 Office Action dated Feb. 13, 2020; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 8 pages.

(56)

References Cited

OTHER PUBLICATIONS

Notice of Allowance dated Jul. 22, 2020; U.S. Appl. No. 16/384,478, filed Apr. 15, 2019; 8 pages.

Office Action dated Jul. 2, 2021; U.S. Appl. No. 17/097,907, filed Nov. 13, 2020; 22 pages.

Notice of Allowance dated Nov. 24, 2021; U.S. Appl. No. 17/097,907, filed Nov. 13, 2020; 8 pages.

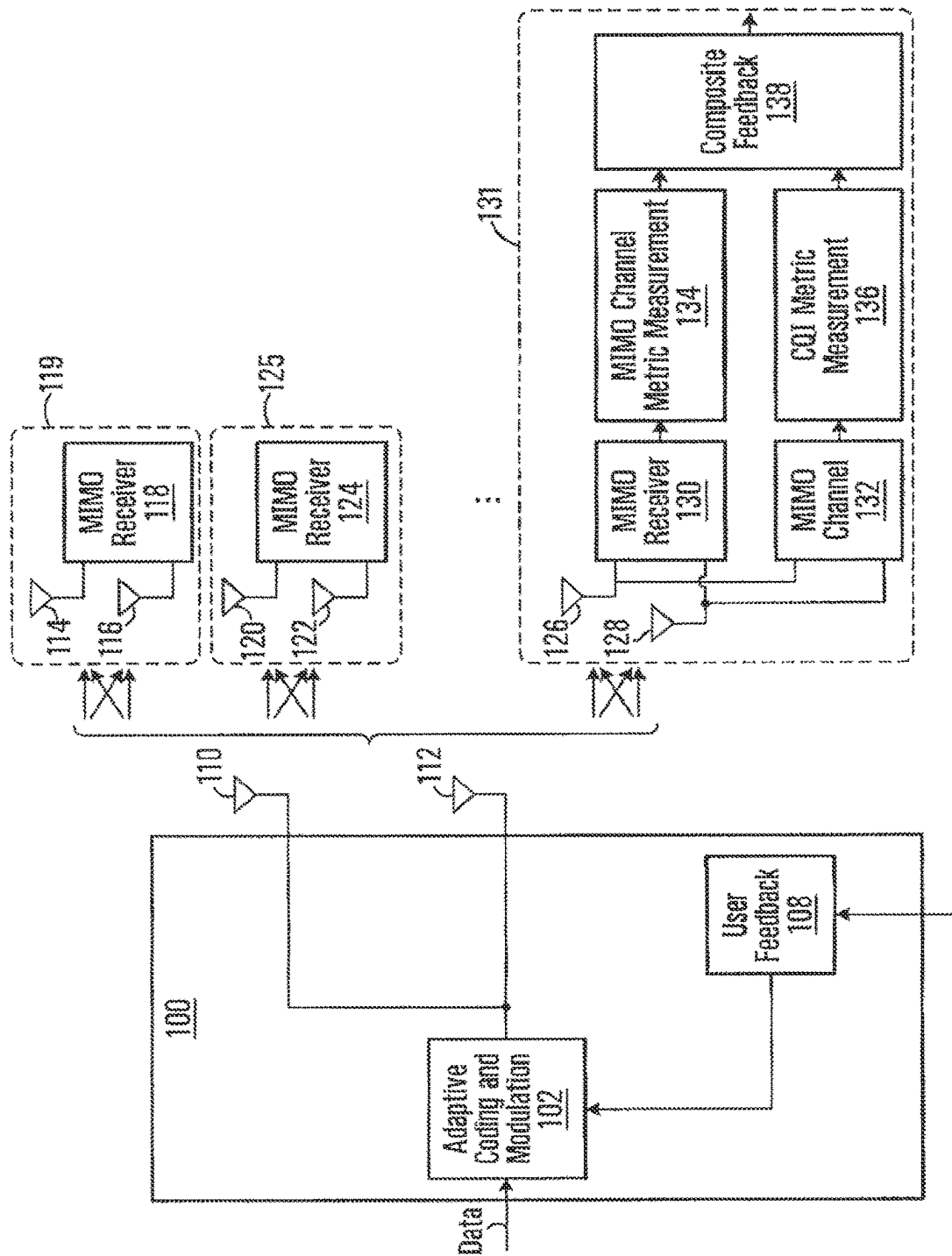
Office Action dated Jan. 18, 2023; U.S. Appl. No. 17/678,456, filed Feb. 23, 2022; 30 pages.

Notice of Allowance dated May 11, 2023; U.S. Appl. No. 17/678,456, filed Feb. 23, 2022; 8 pages.

PCT International Search Report; Application No. PCT/CA2006/001665; Jan. 24, 2007; 2 pages.

PCT Written Opinion of the International Searching Authority; Application No. PCT/CA2006/001665; Jan. 24, 2007; 6 pages.

* cited by examiner



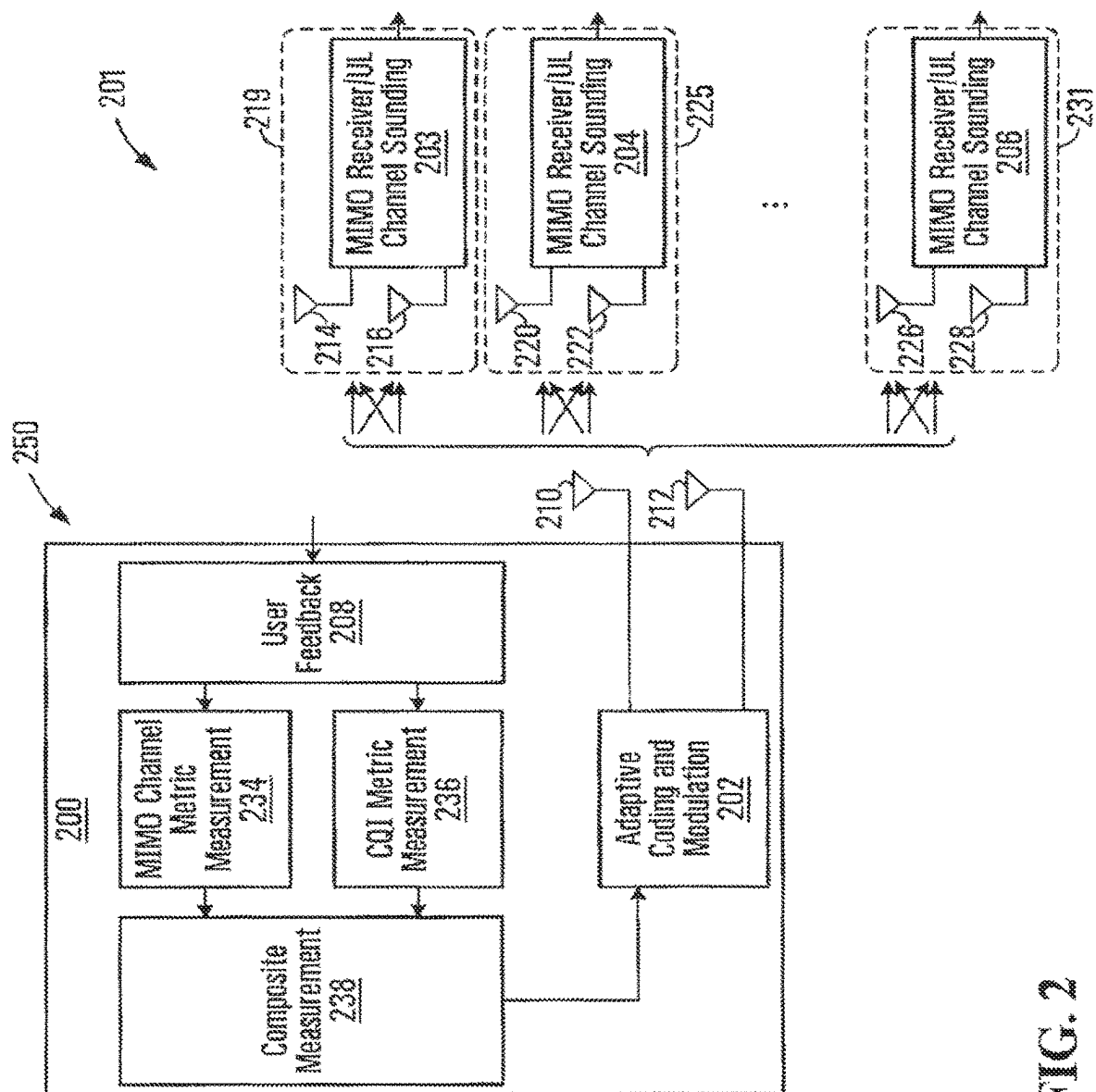


FIG. 2

CQI	1	2	3	4	5	6	7	8	9	10
FEC Code	1	2	3	4	5	3	4	5	3	4
Modulation	1	1	1	1	1	1	1	1	2	2
STC Code	1	1	1	1	1	2	2	2	2	2

FIG. 3A

FEC	Index	1	2	3	4	5
	Codes Rate	R=1/5	R=1/3	R=1/2	R=2/3	R=4/5
Modulation	Index	1	2	3		
	Constellation	4-QAM	16-QAM	64-QAM		
STC	Index	1	2			
	Modes	STTD	BLAST			

FIG. 3B

CQI	1	2	3	4	5	6	7	8	9	10
FEC Code	1	2	3	4	5	3	4	5	3	4
Modulation	1	1	1	1	1	1	1	1	2	2

FIG. 4A

MIMO	1	2	3	4	5	6	7	8	9	10
MIMO Format	1	2	3	4	5	3	4	5	3	4

FIG. 4B

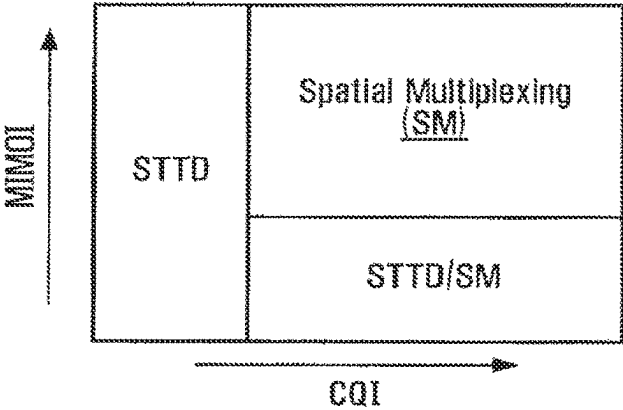


FIG. 5

Impact on Max Capacity Scheduling
(SM 2x2 MLD, 16 UEs, ITU-VA, Turbo R=1/2, 10MHz 1024-FFT, Single Band)

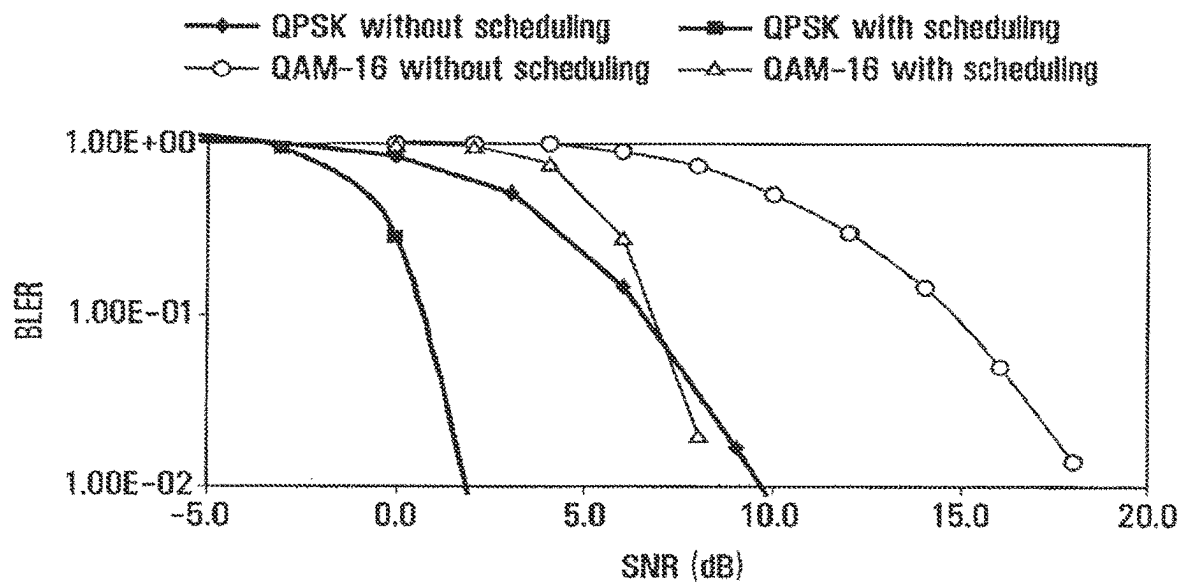
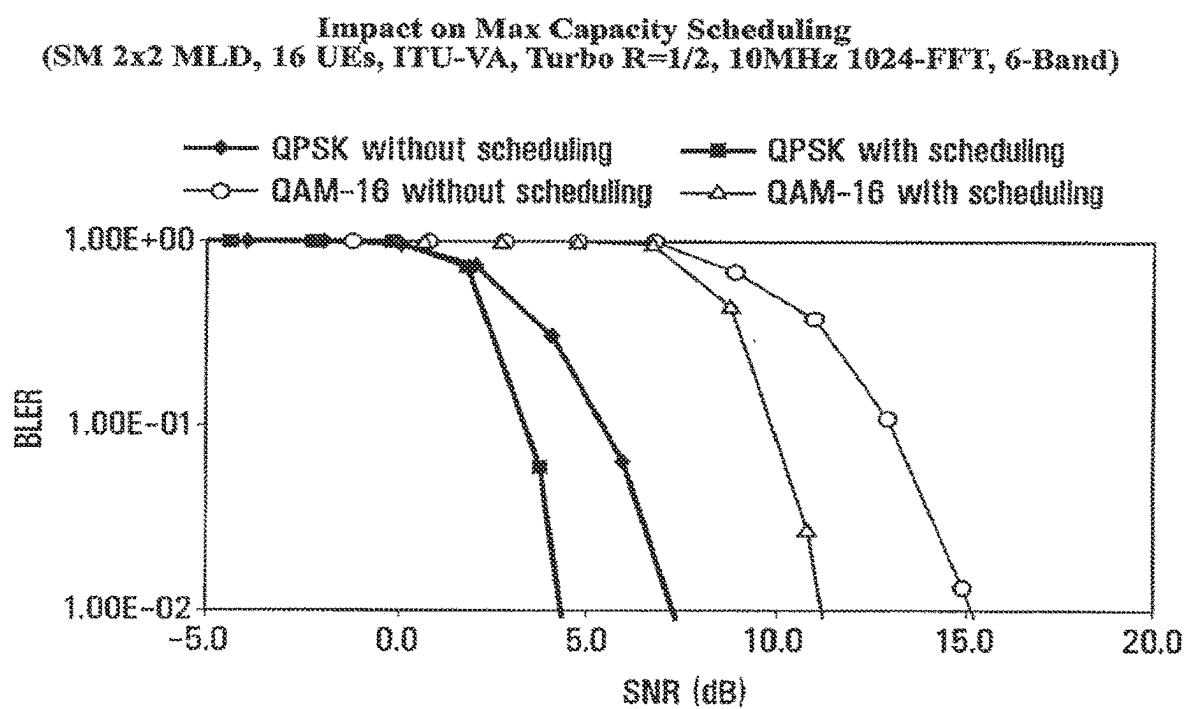


FIG. 6

**FIG. 7**

Minimum Conditional Number Scheduling
(SM 2x2 MLD, 16 UEs, ITU-VA, Turbo R=1/2, 10MHz 1024-FFT, 2-Band)

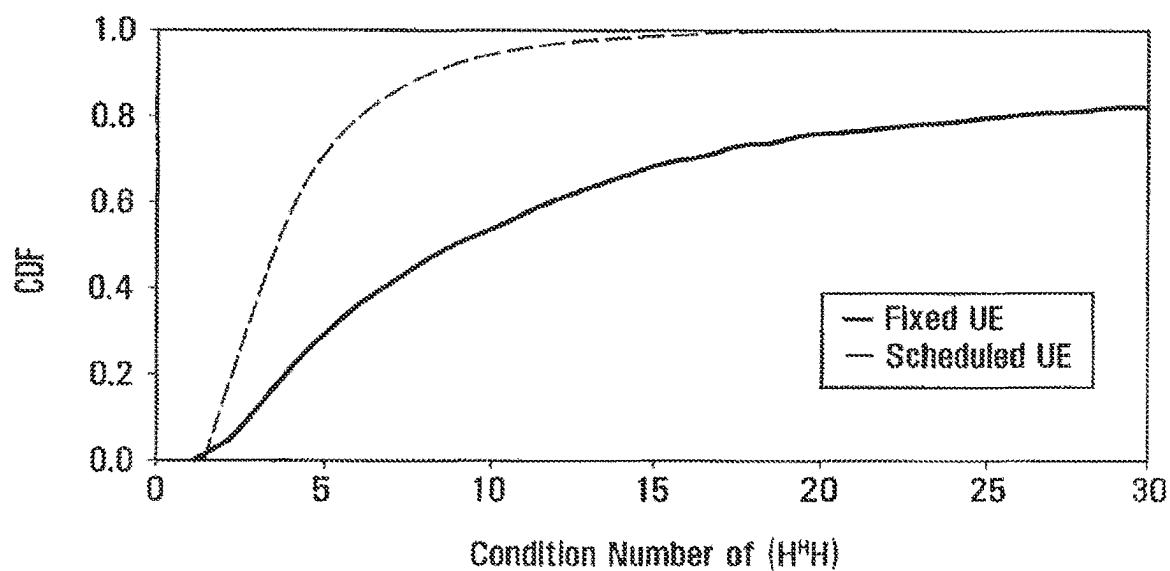


FIG. 8A

Minimum Conditional Number Scheduling
(SM 2x2 MLD, 16 UEs, ITU-VA, Turbo R=1/2, 10MHz 1024-FFT, 6-Band)

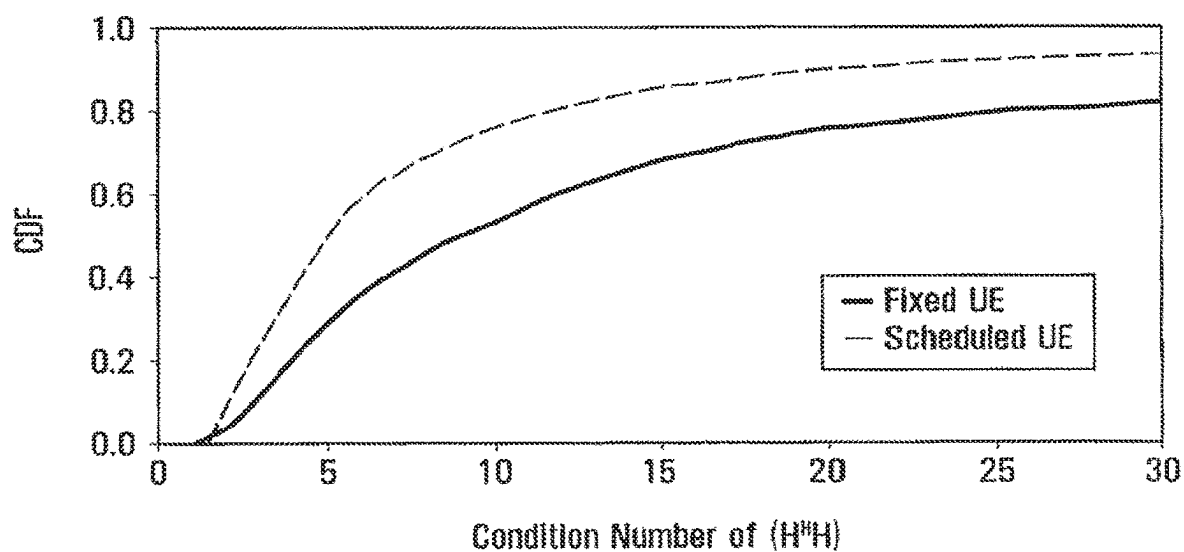


FIG. 8B

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MULTI-USER MIMO SYSTEMS AND METHODS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 17/678,456 filed Feb. 23, 2022 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 17/097,907 filed Nov. 13, 2020 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 16/384,478 filed Apr. 15, 2019, now U.S. Pat. No. 10,856,305 issued on Dec. 1, 2020 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 15/881,575 filed on Jan. 26, 2018, now U.S. Pat. No. 10,306,658 issued on May 28, 2019 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 15/360,600 filed on Nov. 23, 2016, now U.S. Pat. No. 9,918,328 issued on Mar. 13, 2018 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 14/832,758 filed on Aug. 21, 2015, now U.S. Pat. No. 9,538,408 issued on Jan. 3, 2017 entitled “Multi-User MIMO Systems and Methods”, which is continuation of U.S. patent application Ser. No. 13/971,534 filed on Aug. 20, 2013, now U.S. Pat. No. 9,301,174 issued on Mar. 29, 2016 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 13/608,234 filed on Sep. 10, 2012, now U.S. Pat. No. 8,611,454 issued on Dec. 17, 2013 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 13/251,394 filed on Oct. 3, 2011, now U.S. Pat. No. 8,284,852 issued on Oct. 9, 2012 entitled “Multi-User MIMO Systems and Methods”, which is a continuation of U.S. patent application Ser. No. 12/089,938 filed on Apr. 11, 2008, now U.S. Pat. No. 8,054,898 issued on Nov. 8, 2011 entitled “Multi-User MIMO Systems and Methods”, which is a filing under 35 U.S.C. § 371 of International Application No. PCT/CA2006/001665 filed Oct. 12, 2006 entitled “Multi-User MIMO Systems and Methods”, claiming priority to U.S. Provisional Application No. 60/725,951 filed Oct. 12, 2005, all of which are incorporated by reference herein as if reproduced in their entirety.

TECHNICAL FIELD

The present application relates generally to communication systems in general, and, more specifically, to MIMO (multiple-input multiple-output) communication systems.

BACKGROUND

In a MIMO communication system, a transmitter transmits data through multiple transmitting antenna (N_T) and a receiver receives data through multiple receiving antenna (N_R). The binary data to be transmitted is usually divided between the transmitting antennae. Each receiving antenna receives data from all the transmitting antennae, so if there are M transmitting antennae and N receiving antennae, then the signal will propagate over $M \times N$ channels, each of which has its own channel response.

MIMO wireless communication systems are advantageous in that they enable the capacity of the wireless link between the transmitter and receiver to be improved compared with previous systems in the respect that higher data rates can be obtained. The multipath rich environment

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enables multiple orthogonal channels to be generated between the transmitter and receiver. Data can then be transmitted over the air in parallel over those channels, simultaneously and using the same bandwidth. Consequently, higher spectral efficiencies are achieved than with non-MIMO systems.

SUMMARY

In some aspects of the present disclosure, a base station in a multi-user MIMO system selects a transmission method on the basis of feedback information received from a plurality of receivers.

In some aspects, the base station assigns a data rate and a MIMO mode suited to the channel quality for that user.

In some aspects, the present disclosure includes systems and methods which may compute MIMO channel metrics.

In some aspects, the present disclosure includes systems and methods which may include MIMO mode selection.

In some aspects, the present disclosure includes systems and methods which may assign/schedule MIMO user transmission and associated formats in order to maximize MIMO communication capacity.

In some aspects, the present disclosure includes systems and methods which may be used in conjunction with OFDM sub-channels.

In some aspects, the present disclosure includes systems and methods which use uplink (UL) channel sounding where MIMO matrices may be calculated on the transmission side.

According to one broad aspect of the present disclosure, there is provided a method comprising: i) generating a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) channel indication, the MIMO channel indication indicating if the MIMO channel is orthogonal; and ii) transmitting a composite metric based on the MIMO channel indication and the CQI.

According to another broad aspect of the present disclosure, there is provided a transceiver system comprising: i) a generator configured to generate a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) channel indication, the MIMO channel indication indicating if the MIMO channel is orthogonal; and ii) a transmitter configured to transmit a composite metric based on the MIMO channel indication and the CQI.

According to still another broad aspect of the present disclosure, there is provided a method comprising: i) generating a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) mode indication, the MIMO mode indication indicating a MIMO mode; and ii) transmitting a composite metric based on the MIMO mode indication and the CQI.

According to yet another broad aspect of the present disclosure, there is provided a transceiver system comprising: i) a generator configured to generate a channel quality indicator (CQI) and a multiple-input multiple-output (MIMO) mode indication, the MIMO mode indication indicating a MIMO mode; and ii) a transmitter configured to transmit a composite metric based on the MIMO mode indication and the CQI.

Other aspects and features of the present disclosure will become apparent to those ordinarily skilled in the art upon review of the following description of the specific embodiments of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will now be described in greater detail with reference to the accompanying diagrams, in which

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FIG. 1 is a block diagram of a multi-user MIMO system in accordance with an embodiment of the present disclosure where channel measurements are calculated on the receive side;

FIG. 2 is a block diagram of a multi-user MIMO system in accordance with an embodiment of present disclosure where channel measurements are calculated on the transmission side;

FIG. 3A illustrates a lookup table for an embodiment of the present disclosure in which a Forward Error Correction (FEC) code, a modulation type and a MIMO mode are selected based on a Channel Quality Indicator (CQI);

FIG. 3B illustrates a key table for FIGS. 3A and 4A;

FIG. 4A illustrates a lookup table for another embodiment of the present disclosure in which a Forward Error Correction (FEC) code and a modulation type are selected based on a CQI;

FIG. 4B illustrates a lookup table for a MIMO mode based on a MIMO indicator (MIMO);

FIG. 5 is a diagram illustrating a graphical lookup for a MIMO mode based on a MIMO and a CQI;

FIG. 6 is a plot of BLER (Block Error Rate) versus Signal-to-Noise Ratio (SNR) for an embodiment of the present disclosure;

FIG. 7 is a plot of BLER (Block Error Rate) versus SNR for another embodiment of the present disclosure;

FIG. 8A is a plot of Cumulative Distribution Function (CDF) versus condition number of $H^H H$ for an embodiment of the present disclosure; and

FIG. 8B is a plot of CDH versus $H^H H$ for another embodiment of the present disclosure.

DETAILED DESCRIPTION

According to embodiments of the present disclosure, systems and methods are provided which enhance the performance of communication channels in a communication system, to thereby improve, for example, the transmission performance of multi-user MIMO communication systems.

In multi-user MIMO systems, a multi-data stream transmitter at a base transceiver station (BTS) that provides communication services for a coverage area or cell in a wireless communication system transmits communication signals to a plurality of user terminals via multiple antennas. User terminals are also commonly referred to as MIMO receivers, user equipment (UE), communication devices, and mobile stations, for instance. At a MIMO receiver side, multiple receive antennas are employed for each user.

FIG. 1 is a block diagram of a multi-user MIMO system in accordance with one embodiment of the present disclosure. On the transmit side, the system of FIG. 1 includes a BTS 100 with an Adaptive Coding and Modulation Module 102, user feedback module 108, and a pair of antennas 110, 112. On the receive side, the system of FIG. 1 includes one or more user terminals 119, 125, 131 (three shown in the illustrated example) include respective MIMO receivers 118, 124, 130 each having a pair of antennas 114, 116, 120, 122, and 126, 128 respectively.

In user terminal 131, antennas 126, 128 are both connected to the MIMO receiver 130 and to a MIMO channel module 132. MIMO channel module 132 represents the real world radio propagation channel. MIMO receiver 130 is connected to MIMO channel metric measurement module 134. MIMO channel module 132 is connected to CQI metric measurement module 136. Both MIMO channel metric measurement module 134 and CQI metric measurement module 136 are connected to composite feedback module

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138. Composite feedback module 138 forms part of the feedback path from the receive side to the transmit side. Information regarding the MIMO channel metric and the CQI metric is transmitted from MIMO channel metric measurement module 134 and CQI metric measurement module 136 respectively to composite feedback module 138 which incorporates one or more lookup tables to determine a MIMO mode and data rate. The MIMO mode and data rate are fed back by composite feedback module 138 to BTS 100 by any convenient communications method, which may or may not comprise wireless communications.

Each of user terminal 119 and user terminal 125 also include channel measurement modules as well (i.e. each have their own modules equivalent to MIMO channel module 132, MIMO channel metric measurement module 134, CQI metric measurement module 136, and composite feedback module 138). These modules, which are connected to each of MIMO receiver 118 and MIMO receiver 124, have been intentionally omitted to simplify FIG. 1.

The system of FIG. 1 operates as follows. Pilot data is input into adaptive coding and modulation module 102 where such pilot data is converted into communication signals which may then be transmitted via the antennas 110, 112 from the BTS 100 to user terminals 119, 125 and 131. In some embodiments, the pilots are inserted on each antenna in a manner that makes them distinguishable at a receiver. For example, for OFDM implementations, a respective set of sub-carrier and OFDM symbol durations can be employed for each antenna.

At MIMO receivers 118, 124, and 130, each of the antennas 114, 116, 120, 122, and 126, 128 receive the pilot signals transmitted from the antennas 110, 112. MIMO receiver 130 processes the received signals to produce separated layer signals which are fed to MIMO channel metric measurement module 134. The MIMO channel measurement metric measurement module 134 processes the received pilot data having regard to knowledge of what the transmitted pilot data was, and produces a MIMO channel metric. Specific examples of calculations which may be performed to assess a MIMO channel metric are described below. MIMO channel module 132 processes the received signal to produce MIMO channel state information which is fed to the CQI measurement module 136. The CQI metric measurement module processes the received pilot data having regard to knowledge of what the transmitted pilot data was, produces a CQI metric. CQI metrics are well known and may for example include CINR (carrier to interference and noise ratio), and the rank of the MIMO channel.

The CQI metric is used as a basis for selecting a particular coding and modulation. BTS 100 can adjust the modulation order and/or coding rate in accordance with the CQI metric. More particularly, the data transmission rate can be increased, decreased, maintained at a constant level, or reduced to 0 bits/s. In a particular example, the CQI is CINR as indicated above, and each range of CINR is associated with a respective adaptive coding and modulation.

In some embodiments, MIMO receivers 118, 124 and 130 track the channel quality via the pilot symbols received and accumulate these quality measurements over a period of time to produce the CQI.

In some embodiments, the feedback from user terminals 119, 125, 131 may also include information identifying the receiver's MIMO capability. For example, this might indicate a number of receive antennas, or the rank of the MIMO channel.

The MIMO channel metric is used to select a MIMO transmission mode to be used for transmitting to a particular

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user terminal. The particular MIMO modes that are available are selected on an implementation specific basis. Four examples of MIMO modes include beamforming, BLAST, space-time transmit diversity (STTD), and spatial multiplex, though the present disclosure is in no way limited to these MIMO modes and is in fact applicable to all possible space-time mapping.

Those skilled in the art will appreciate that MIMO channel metric measurement and CQI metric measurement may be performed by a digital signal processor (DSP) or a general-purpose processor adapted to execute signal processing software, for example. Various techniques for determining such metric measurements will be apparent to those skilled in the art.

Both the MIMO channel metric and the CQI is transmitted to composite feedback module 138 where one or more lookup tables may be used to determine a composite metric used by BTS 100 to select a MIMO mode and data rate. As used herein, "composite" can be equated to the "overall" quality of the channel matrix. The lookup carried out by composite feedback module 138 is used for two purposes: (i) User terminal pairing, i.e. scheduling. The more orthogonal the channel, the larger the MIMO capacity; and (ii) together with SNR, the lookup is used for MIMO mode and coding and modulation selection. With a higher SNR and composite metric, spatial multiplexing and higher modulation and coding rates may be selected. With a lower SNR and composite metric, transmit and lower modulation and coding rates may be selected.

Note that the composite metric does not affect modulation and coding rates selection in transmit diversity, but it affects modulation and coding rates selection in spatial multiplexing. This is because when the composite metric is low, more inter-layer interference will occur, and hence only lower modulation and coding rates are to be used.

The composite metric is then transmitted by composite feedback module 138 to BTS 100 through user feedback module 108. With the composite metric received from composite feedback module 138, a scheduler which forms part of BTS 100 determines a MIMO transmission mode and a modulation and coding to be used for each MIMO receiver. In some embodiments, the BTS 100 indicates the transmission format to each MIMO receiver.

In some embodiments, a two bit composite metric is used, with one bit of the composite metric being used to indicate the CQI, and one bit of the composite metric to indicate the MIMO mode, e.g. transmit diversity or spatial multiplexing. In the spatial multiplexing mode, one additional bit can be used to indicate if the MIMO channel is orthogonal.

FIG. 2 is an alternative embodiment to that illustrated in FIG. 1. In this embodiment, channel metrics are measured at the transmit side 250 rather than the receiver side 201. On the transmit side 250, the system of FIG. 2 includes a BTS 200 with an Adaptive Coding and Modulation Module 202, user feedback module 208, and a pair of antennas 210, 212. On the receive side 201, the system of FIG. 2 includes one or more user terminals 219, 225, 231 (three shown in the illustrated example) include respective MIMO receivers 203, 204, 206 each having a pair of antennas 214, 216, 220, 222, and 226, 228 respectively.

In user terminals 219, 225, and 231, antennas 214, 216, 220, 222, and 226, 228 respectively are connected to MIMO receivers 203, 204, and 206 which each perform UL channel sounding. In the case of Time Division Duplex (TDD), channel sounding is used to allow BTS 200 to perform channel measurements at the transmit side 250 rather than the receiver side 201. Information received from MIMO

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receivers 203, 204, and 206 is fed back through a feedback control channel to user feedback module 208 at BTS 200 by any convenient communications method, which may or may not comprise wireless communications.

User feedback module 208 is connected to both MIMO channel metric measurement module 234 and CQI metric measurement module 236. Both MIMO channel metric measurement module 234 and CQI metric measurement module 236 are connected to composite feedback module 238. Composite feedback module 238 is connected to adaptive coding and modulation module 202.

Except for the fact that channel measurements are performed at the transmit side 250 rather than the receive side 201, the operation of the system of FIG. 2 is otherwise similar to the operation of the system of FIG. 1. The main difference is that through channel sounding, user terminals 219, 225, and 231 pass the burden of channel measurements and processing to BTS 200.

Of course, the systems of FIGS. 1 and 2 are only two illustrative examples of systems in which the present disclosure may be implemented. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to systems having other dimensions will be apparent to those skilled in the art. In particular, the number of user terminals that will be present in a given implementation may differ, and may vary over time if they are mobile. The number of antennas on the base station and user terminals is two in the illustrated example. More generally, any number, two or more, of antennas can be employed such that MIMO communications are possible, though the number of receive antennas must be greater than to equal to the number of data streams (i.e. layers) being transmitted. Both the base station and user terminals include functionality not shown as would be understood to one of skill in the art. Separate components are shown for each of the MIMO channel 132, MIMO channel metric measurement module 134, 234, the CQI metric measurement module 136, 236, and the composite feedback module 138, 238. More generally, the functions provided by these modules may be combined in one or more functional elements, and these may be implemented in one or a combination of software, hardware, firmware etc.

A MIMO system can be expressed as

$$\bar{y} = H \bar{s} + \bar{\eta},$$

where

$\bar{y} = [y_1 \ y_2 \ \dots \ y_N]^T$ is a vector of communication signals received at a receiver;

$\bar{s} = [s_1 \ s_2 \ \dots \ s_M]^T$ is a vector of communication signals transmitted by a transmitter;

$\bar{\eta} = [\eta_1 \ \eta_2 \ \dots \ \eta_N]^T$ is a vector of noise components affecting the transmitted communication signals;

$$H = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1M} \\ h_{21} & h_{22} & \dots & h_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ h_{N1} & h_{N2} & \dots & h_{NM} \end{bmatrix}$$

is a channel matrix of communication channel attenuation factors;

N is a number of antennas at the receiver; and
M is a number of antennas at the transmitter.

For a [2Tx, 2Rx] MIMO channel,

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$$H = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

The eigenvalue of $H^H H$ are $\lambda_{\max}$, $\lambda_{\min}$. There are several scheduling approaches, including orthogonality and capacity. These approaches are for user terminal pairing only.

Where it is desired that scheduling by a BTS (such as BTS **100** and BTS **200** in FIGS. **1** and **2** respectively) be provided on the basis of maximum orthogonality, the following MIMO channel metric will be computed by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\max \left\{ \frac{\det(H^H H)}{\text{trace}(H^H H)} \right\}$$

The larger the metric, the more orthogonal is the channel.

For a maximum orthogonality decomposition scheduling scheme,

$$\min \left\| \begin{bmatrix} h_{11} & h_{21} \end{bmatrix} \begin{bmatrix} * \\ h_{12} \\ * \\ h_{22} \end{bmatrix} \right\| = 0$$

In this case, the channel is completely orthogonal, yielding two separate spatial channels, with channel attenuation factors being $\sqrt{|h_{11}|^2 + |h_{21}|^2}$ and $\sqrt{|h_{12}|^2 + |h_{22}|^2}$ respectively.

For scheduling based on a best conditional number of MIMO channel scheme, the following channel metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\rho = \lambda_{\max} / \lambda_{\min} = -1$$

In this case, an advanced receiver (maximum likelihood detection) and/or a simplified receiver can be employed.

For scheduling based on maximum capacity, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\det(H^H H)$$

Maximum capacity scheduling is also maximum CQI scheduling.

For scheduling based on maximum orthogonality for several MIMO channels,

$$H_1 = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

$$H_2 = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\max \left\{ \sum_{i=1}^2 \frac{\det(H_i^H H_i)}{\text{trace}(H_i^H H_i)} \right\}$$

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For scheduling based on orthogonality capacity for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\max \left\{ \sum_{i=1}^2 \det(H_i^H H_i) \right\}$$

For scheduling based on a combined conditional number for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\min \left\{ \sum_{i=1}^2 \rho_i \right\}$$

For scheduling based on a SNR weighted maximum orthogonality scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\max \left\{ \sum_{i=1}^2 SNR_i \frac{\det(H_i^H H_i)}{\text{trace}(H_i^H H_i)} \right\}$$

For scheduling based on a SNR weighted capacity scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\max \left\{ \sum_{i=1}^2 SNR_i \det(H_i^H H_i) \right\}$$

For scheduling based on a SNR weighted combined conditional number scheme for several MIMO channels, the following metric will be calculated by, for example, MIMO channel metric measurement module **134** in FIG. **1**:

$$\min \left\{ \sum_{i=1}^2 SNR_i \rho_i \right\}$$

FIG. **3A** illustrates a table setting out one representative example of a lookup table for selecting coding modulation and MIMO modes based CQI which can be used in accordance with one embodiment of the present disclosure. The table in FIG. **3B** provides a key for the table in FIG. **3A**.

In the table of FIG. **3A**, Row 1 lists possible CQIs, which in this case would be from 1 to 10. Each CQI has an associated FEC code, an associated modulation, and an associated MIMO mode, in this case an STC code. In the particular example, illustrated, there are five available FEC codes, three available modulations, and two available STC codes.

The key shown in FIG. **3B** indicates the code rates $1/5$, $1/3$, $1/2$, $2/3$, $4/5$ associated with the five available FEC codes, indicates the modulation constellations 4-QAM, 16-QAM 64-QAM associated with the three modulations, and indicates the STC modes STTD and BLAST associated with the two available STC modes.

FIG. 3A shows how CQI, together with STC code rate, determines the proper code and modulation set. The table in FIG. 3A is of course just one possible example of a lookup table. The particular MIMO modes, FEC codes, and modulations supported will vary on an implementation specific basis. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to others possible lookup tables will be apparent to those skilled in the art. For example, other forms of lookup tables could be employed which include other standard modulation schemes such as Phase-shift keying (PSK), and other forms of MIMO modes such as beamforming.

FIGS. 4A and 4B illustrate a lookup table in which the CQI and MIMO are fed back in separate feedback components. In this case, the MIMO mode is determined by both CQI and MIMO.

FIG. 4A sets out one example for coding modulation based on CQI. Row 1 lists possible CQIs which may be calculated by CQI metric measurement module 136, from 1 to 10. Row 2 lists Forward Error Correction Codes (FEC), which in this case are from 1 to 5. Row 3 lists three possible forms of modulation which can be implemented in accordance with this lookup table, namely 4-QAM, 16-QAM, and 64-QAM. Reference may be had to the table in FIG. 3B which provides a key for the table in FIG. 4A.

FIG. 4B is a lookup table for MIMO modes based on a MIMO indicator (MIMO). MIMO indicator may be calculated by MIMO channel metric measurement module 134 shown in FIG. 1. In this case, Row 1 lists possible MIMO indicators from 1 to 10 which are selected based on the result of the channel metric calculations. Row 2 lists possible MIMO formats (which could represent, e.g., BLAST, STTD, beamforming, spatial multiplexing, etc.) from 1 to 5.

FIGS. 4A and 4B show that the higher the CQI/MIMO, the higher is the modulation order and FEC coding rates. There are two situations where a composite metric of both CQI and MIMO may be used: the first is MIMO mode selection, because when MIMO is low, SM may not be used even with a large CQI. FIG. 8 shows that at low MIMO, in the high CQI region, the use of STTD or SM as a MIMO mode depends on MIMO. However, in the same CQI region, when MIMO is larger, SM is selected. Another situation is modulation and code set selection in SM, because in SM, MIMO indicates inter-layer interference, and hence indicates the performance of the channel. Given the same CQI, modulation and code set will depend on the MIMO indicator. In other words, modulation and code set will be determined by two parameters: CQI and MIMO.

As with the table in FIG. 3A, the tables in FIGS. 4A and 4B are merely illustrative examples of lookup tables which could be used in accordance with the present disclosure. Persons skilled in the art will appreciate that other combinations of parameters can be employed.

FIG. 5 is a diagram illustrating another example of a lookup for MIMO modes based on a MIMO indicator and a CQI. As noted above, a MIMO may be ascertained by calculations performed by MIMO channel metric measurement module 134 shown in FIG. 1 and CQI may be calculated by CQI metric measurement module 136 also shown in FIG. 1.

In FIG. 5, three sections are shown, a first section labelled "STTD", a second section labelled "spatial multiplexing (SM)", and a third section labelled "STTD/SM". According to the lookup of FIG. 5, for low CQIs (i.e. low channel quality), regardless of the MIMO indicator, STTD should be chosen as the MIMO format. For high MIMO indicators and high CQIs, the lookup of FIG. 5 indicates that spatial

multiplexing should be chosen as the MIMO format. For a low MIMO indicator and high CQI, either of STTD and spatial multiplexing can be selected as the MIMO format.

For the example of FIG. 5, a CQICH (channel quality indicator channel) can be used to feedback coding/modulation information and/or selection, and a single bit can be used to flag the MIMO mode. One of '0' or '1' can be used for STTD, and the other of '0' or '1' can be for SM.

FIG. 5 is of course just one possible example of a lookup in which the present disclosure may be implemented. The present disclosure is in no way limited thereto. Extension of the principles of the present disclosure to other lookup diagrams will be apparent to those skilled in the art.

FIG. 6 is a plot of BLER (Block Error Rate) versus SNR (Signal-to-Noise Ratio) for an embodiment of the present disclosure. FIG. 6 shows the impact on maximum capacity scheduling. FIG. 7 is a plot of BLER (Block Error Rate) versus SNR (Signal-to-Noise Ratio) for another embodiment of the present disclosure. FIG. 7 shows the impact on maximum capacity scheduling. Both FIGS. 6 and 7 shows two sets of curves, namely, "without scheduling" and "with scheduling". More particularly, "without scheduling" means "random scheduling", and "with scheduling" means "orthogonality based scheduling".

FIG. 8A is a plot of Cumulative Distribution Function (CDF) versus condition number of $H^H H$ for an embodiment of the present disclosure. $H^H H$ means Hermitian transform, i.e. conjugate transposition. FIG. 8A presents the impact on minimum conditional number scheduling. FIG. 8A shows the CDF of the composite metric when random pairing is used (the solid line) and when the orthogonality based pairing is used.

FIG. 8B is a plot of CDH versus (HHH) for another embodiment of the present disclosure. FIG. 8B presents the impact on minimum conditional number scheduling.

Numerous modifications and variations of the present disclosure are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the present disclosure may be practised otherwise than as specifically described herein.

What is claimed is:

1. A base station comprising:

a transceiver; and

a processor operationally coupled to the transceiver;

wherein the processor is configured to:

receive, via the transceiver, channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and transmit, via the transceiver, a MIMO transmission using a number of transmission antennas based on the CSI feedback, wherein the number of transmission antennas is based on the rank and an adaptive coding and modulation type for the transmission selected is based on the CQI.

2. The base station of claim 1, wherein a data transmission rate of the MIMO transmission is based on the selected adaptive coding and modulation type.

3. The base station of claim 2, wherein data transmission rates of subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the received CSI feedback.

4. The base station of claim 1, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.

5. The base station of claim 1, wherein the MIMO transmission includes data and pilot signals, wherein the

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pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.

6. A method for use by a base station, the method comprising:

receiving channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and

transmitting a MIMO transmission using a number of transmission antennas based on the CSI feedback, wherein the number of transmission antennas is based on the rank and an adaptive coding and modulation type for the transmission is selected based on the CQI.

7. The method of claim 6, wherein a data transmission rate of the MIMO transmission is based on the selected adaptive coding and modulation type.

8. The method of claim 7, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the received CSI feedback.

9. The method of claim 6, wherein the MIMO transmission is via an Orthogonal Frequency Division Multiplexing (OFDM) transmission.

10. The method of claim 6, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.

11. A user equipment (UE) comprising:

a transceiver;

a processor, operatively coupled to the transceiver, wherein the transceiver and the processor are configured to:

transmit, via the transceiver, channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and

receive, via the transceiver, a MIMO transmission based on the CSI feedback, wherein the rank is used for selecting a number of transmission antennas for the MIMO transmission and the CQI is used for selecting an adaptive coding and modulation type for the MIMO transmission.

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12. The UE of claim 11, wherein a data transmission rate of the MIMO transmission is based on the CSI feedback.

13. The UE of claim 12, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the transmitted CSI feedback.

14. The UE of claim 11, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.

15. The UE of claim 11, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.

16. A method for use by a user equipment (UE), the method comprising:

transmitting channel state information (CSI) feedback that includes any one or a combination of a channel quality indicator (CQI) and a rank of a multiple-input multiple-output (MIMO) channel; and

receiving a MIMO transmission based on the CSI feedback, wherein the rank is used for selecting a number of transmission antennas for the MIMO transmission and the CQI is used for selecting an adaptive coding and modulation type for the MIMO transmission.

17. The method of claim 16, wherein a data transmission rate of the MIMO transmission is based on the CSI feedback.

18. The method of claim 17, wherein data transmission rates of the subsequent MIMO transmissions can be increased, decreased, or maintained at a constant level based on the transmitted CSI feedback.

19. The method of claim 16, wherein the MIMO transmission is an Orthogonal Frequency Division Multiplexing (OFDM) transmission.

20. The method of claim 16, wherein the MIMO transmission includes data and pilot signals, wherein the pilot signals are inserted on each antenna in a set of sub-carriers and OFDM symbol durations.

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